

William Marella

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Senior data scientist specializing in causal inference, experimentation, machine learning, and applied AI. Led analytical strategy for a \$10M ARPA-H program and invented a patent-pending epigenetic aging algorithm commercially licensed to TruDiagnostic.

EDUCATION

University of Cambridge

MPhil. Population Health Sciences, Specialization in Health Data Science

November 2024

Brown University

B.A. Biology, GPA: 3.96

May 2023

WORK EXPERIENCE

Senior Data Scientist

Columbia University, Geroscience Computational Core

New York, New York

September 2024 - Present

- Lead analytics and engineering for the Core, translating ambiguous scientific questions and collaborator needs into causal-inference pipelines, ML workflows, statistical software, and decision-ready analyses for technical and nontechnical stakeholders.

ARPA-H FAST Grant | R, Python, Causal Inference, ML

- Led design and execution of causal inference and ML strategy for a **\$10M ARPA-H grant**; architected collaborator-facing R/Python pipelines used by clinical and industry partners including **Novo Nordisk and Lilly**.
- Translated grant aims and scientific questions into reproducible, portable analytical workflows: treatment-effect estimation and mediation analysis for high-dimensional clinical and omics data, and ML-based prediction of clinical endpoints from repeated-measures omics.

QueryGaP | Python, SQL, PostgreSQL, TypeScript, Next.js, MCP | [Website](#), [MCP Repository](#)

- Built and deployed an **LLM-powered search platform** for dbGaP and UK Biobank documentation, reducing phenotype-variable discovery from hours to minutes and gaining adoption at Columbia and external institutions.
- Open-sourced and deployed QueryGaP as a **hosted remote MCP server** enabling ChatGPT, Claude, Codex, and other MCP clients to retrieve grounded, source-linked records across dbGaP, UK Biobank, and All of Us.

FraminghamPACE | R, Hierarchical Modeling, ML | [medRxiv](#), [GitHub](#)

- Developed a hierarchical model of the pace of aging and machine learning training methods to create a **state-of-the-art epigenetic aging clock** with improved predictive performance, technical reliability, and responsiveness to intervention.
- First author** of the resulting manuscript, currently in review; **inventor** of the patent-pending algorithm, **commercially licensed** by Columbia University to TruDiagnostic.

Visiting Researcher

University of Cambridge, Inouye Lab

Cambridge, UK

June 2024 - June 2026

- Derived mean-field variational inference and partially collapsed Gibbs sampling algorithms for Latent Factor Allocation, an unsupervised Bayesian machine learning method; benchmarked their accuracy and computational complexity on simulated data designed to resemble UK Biobank data.
- Implemented both algorithms from scratch in NumPy/SciPy, including convergence diagnostics and multithreaded MCMC chain coordination, and released the work as a [public Python library](#).

SELECTED PRESENTATIONS

- NIA Biomarker Network Meeting 2026** | St. Louis, MO | Invited Oral Presentation
FraminghamPACE: A New DNA Methylation Biomarker of the Pace of Aging
- Biomarkers of Aging Consortium 2025** | Boston, MA | Poster Presentation
FraminghamPACE: An Epigenetic Speedometer of the Pace of Aging

SELECTED AI & SOFTWARE PROJECTS

Forecasting Reasoning Length from Model Internals | Python, PyTorch, Transformers, AWS | [Technical Writeup](#)

- Built a **10,000-generation benchmark** to demonstrate that prompt-time hidden states predict an LLM's next sampled output length beyond prompt text. **Reduced mean absolute error by 169 tokens** relative to a strong text-only ensemble on 1,000 untouched Qwen3-4B/OASST2 prompts

- Engineered an **interruption-resilient AWS Spot pipeline** on NVIDIA L4s with frequent S3 checkpointing, encrypted storage, automated recovery, and reproducibility safeguards.

Text-to-ASCII Diffusion Model | *Python, PyTorch, Transformers* | [GitHub](#), [Hugging Face](#)

- Text-to-ASCII art generator powered by a **300M-parameter Diffusion Transformer (DiT)** trained with flow matching over a **VAE-compressed latent space** of 32x16 ASCII token grids.
- Trained on Quick, Draw! using NVIDIA A100 GPUs on Modal and published the model weights on Hugging Face.
- Conditioned the model on **CLIP text embeddings** with classifier-free guidance to control prompt adherence; demonstrated generalization to synonyms and near-out-of-distribution prompts unseen during training.

OS-Level AI Agent Sandboxing | *TypeScript, Node.js, Bun, Unix/Linux* | [GitHub](#)

- Forked OpenCode to implement **OS-level isolation for AI agent-generated shell commands**, running tool execution under a restricted Unix user to protect credentials and sensitive files.
- Used Unix permissions as a security boundary independent of model behavior, constraining agent access even when malicious repository content or tool outputs bypass prompt-level safeguards.

Multi-Agent LLM Strategic Reasoning Benchmark | *Python, OpenAI SDK* | [Technical Writeup](#), [GitHub](#)

- Built a **controlled multi-agent benchmark** to test whether LLM failures to reach Nash equilibria reflect cooperative bias or limits in strategic reasoning, varying model scale and inference-time reasoning across increasingly difficult environments.
- Found that frontier agents readily pursued self-interested equilibria, while success in the hardest environment depended sharply on reasoning budget.

CostcoGuessr Web App | *TypeScript, Next.js, Bun, SQL, PostgreSQL* | [Website](#)

- Built and **deployed a full-stack daily price-guessing game** with a TypeScript/Bun backend, Next.js frontend, PostgreSQL database, and image storage on Vercel Blob.
- Scraped and normalized 10,000+ Costco products across 11 countries**, including product metadata, prices, and images.
- Grew organically to **200 daily users**.

Open-Source Contributions: *TypeScript, React, Bun, Node.js* | [OpenCode PR](#), [PostHog PR](#)

- Merged navigation and UI features and fixes into **OpenCode** and **PostHog** codebases.

SKILLS

Languages & Systems: Python, R, SQL, TypeScript, JavaScript, Bash, Unix, Linux

Experimentation & Causal Inference: experimentation, A/B testing, metric design, treatment-effect estimation, mediation analysis, quasi-experimental design, difference-in-differences, propensity score methods, sensitivity analysis, hypothesis testing, power analysis

Statistical Modeling: mixed-effects models, survival analysis, longitudinal modeling, Bayesian inference, hierarchical models, MCMC, variational inference, multiple testing, false discovery rate (FDR), model diagnostics

Machine Learning: NumPy, SciPy, PyTorch, Hugging Face Transformers, scikit-learn, supervised and unsupervised learning, regularized regression, gradient boosting, Optuna, feature selection, model interpretability, NLP, text mining, clustering, dimensionality reduction

AI Systems & Evaluation: LLM applications, retrieval-augmented generation (RAG), semantic search, vector search, Model Context Protocol (MCP), LangChain, LangGraph, AI agents, agentic workflows, prompt engineering, context engineering, LLM evaluation, model behavior analysis, synthetic data generation, multi-agent simulation

Data & Software Engineering: PostgreSQL, pgvector, AWS (EC2, S3, Spot), Snowflake, APIs, data pipelines, web scraping, Docker, Git, GitHub, reproducible analysis, Next.js, React, Node.js, Bun, Vercel, Railway